GRADIA RESEARCH

Interruptible Universes: Evolution Witnesses for Verifiable World Change in Agent Benchmarks

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Contents

Locked claim boundary. This repository demonstrates deterministic harness semantics on five fully synthetic control conditions and 15 scripted-policy episodes. It also admits five synthetic frontier-candidate task editions for deterministic solvability and evaluator sensitivity only. Two additional PRE-RESULTS axes freeze five phase-response and five authority-ladder candidate/control pairs with exact witnesses and mutation probes; no model has run on them. The artifact does not yet establish language-model capability, empirical task difficulty, phase or authority effects, human agreement, real-world validity, training lift, causal attribution, provider parity, or research novelty. Scripted-policy and evaluator-probe results below are engineering checks only. Every unmeasured empirical result is marked **NOT YET MEASURED**.

Abstract

Long-horizon agents act while their worlds continue to change. Evidence can arrive, a policy can be superseded, a source can be retracted, an operator can interrupt, or a runtime can restore from a snapshot. A benchmark may record a final state and an event log yet still be unable to establish whether a declared change was applied at the right action boundary, whether the agent saw the declared projection, or whether restore preserved exactly-once history. This paper studies a narrow evidence object for that problem: an **evolution witness**.

An evolution witness composes a frozen exogenous-event contract with its realized application: before/after material-world roots, an exact model-visible projection, an action boundary, restore generation, and the previous occurrence-chain head. The public artifact implements this object in a small synthetic underwriting universe containing a static control, document addition, policy revision, retraction across restore, and an authority conflict. Its committed reference panel comprises 15 deterministic episodes from three deliberately constructed scripted policies. Those controls pass 5/5, 2/5, and 1/5 episodes respectively and expose the intended judge and failure-taxonomy sensitivity; they are not estimates of model behavior. A separate five-edition candidate suite expands the workload to a four-case queue, six versioned sources, shared exception capacity, chained cutoff events and restore. Its safe-solver admissions and 44 isolated judge probes establish only solvability and executable criterion sensitivity. A PRE-RESULTS corpus additionally freezes two five-case axes: the same revision at five workflow phases and five source-authority rungs at one phase. Its ten seed-paired controls, ten exact witnesses, and 100 isolated probes establish reproducible construction only. The artifact also commits replayable episode receipts, generated reports, an unsigned public-bundle candidate, adversarial integrity tests, and explicit data and rights cards.

The paper preregisters the still-unrun studies required for a scientific claim. The primary study injects one isolated validity mutation per paired receipt and tests whether full witnesses detect failures missed by terminal-state, ordinary-log, milestone, and proof-of-execution-style baselines. A separately pinned live-model panel measures behavioral adaptation, while a blinded human study measures criterion agreement and witness usefulness. The primary endpoint is paired incremental invalid-episode detection; faithful-episode false rejection, localization, replay, abstention, and verification cost are secondary endpoints. The proposed delta is not the invention of dynamic worlds, interruptions, provenance, or execution receipts. It is the benchmark-specific composition of declared world evolution with material application, disclosure, temporal boundary, and restore lineage–and whether that composition provides incremental validity evidence.

1. Introduction

Agent benchmarks are becoming less like question sets and more like operating environments. An agent reads tools, writes state, waits for external events, handles interruptions, and eventually commits an answer or action. This is desirable: realistic work is stateful and open ended. It also creates a measurement problem that a terminal reward cannot solve by itself.

Suppose an environment declares that a policy was revised after the agent's second act. The final database may contain the revised policy. The log may contain a revision event. The agent may nevertheless have received a different message, received it after committing, or continued from a restored snapshot in which the revision fired twice. A correct terminal state cannot prove a correct trajectory, and an ordinary log cannot prove that the logged event corresponds to the material change, visible notice, or boundary the benchmark specification promised.

This distinction matters for both evaluation and training. If an invalid environment episode is scored as an agent failure, a leaderboard measures harness defects. If it is admitted into a training corpus, the reward signal may favor stale evidence, unauthorized instructions, or behaviors that exploit replay bugs. Conversely, an evidence system that rejects valid episodes too readily can make long-horizon evaluation impractical. The empirical question is therefore not whether more logging sounds useful. It is whether a particular evidence composition detects additional invalid episodes without unacceptable false rejection or cost.

The artifact in this repository turns that question into a falsifiable study. It provides a minimal world where all relevant state, events, observations, actions, restores, and judgments can be audited end to end. It deliberately avoids real records, non-public infrastructure, and organization-dependent claims. The synthetic domain supplies understandable state changes; it is not presented as a lending benchmark or policy.

1.1 Research questions

- **RQ1 – Incremental validity.** Does a full evolution witness detect invalid dynamic-world episodes missed by terminal-state comparison, an ordinary event log, action milestones, or a proof-of-execution-style causal baseline?
- **RQ2 – Projection and authority.** Does binding the exact visible projection distinguish application failures, disclosure failures, authority changes, and agent decision failures?
- **RQ3 – Restore semantics.** Does binding restore generation and occurrence-chain lineage detect duplicate, missing, or reordered delivery across snapshot/restore while accepting faithful restores?
- **RQ4 – Behavioral adaptation.** Under a fixed scaffold, how often do identity-bound live-model cells re-read authoritative state and adapt after additions, revisions, retractions, and non-authoritative interruptions?
- **RQ5 – Human utility.** Do evolution witnesses improve blinded reviewers' detection and localization of invalid episodes relative to weaker evidence views?
- **RQ6 – Portability.** Can independently implemented runtimes produce semantically equivalent witness evidence under one frozen provider-neutral contract?

Only the harness-sensitivity portion of these questions has been exercised in the committed reference panel. RQ1-RQ6 as empirical research questions remain **NOT YET MEASURED**.

1.2 Contribution ladder

The paper separates what is implemented, what is measured, and what remains hypothetical.

1. **Implemented evidence object.** A canonical, hash-bound occurrence receipt joins a declared event, realized before/after roots, exact visible projection, action boundary, restore generation, and prior chain head.
2. **Implemented auditable universe.** Five synthetic conditions exercise no change, evidence addition, policy supersession, retraction across restore, and an unauthorized instruction over one common tool contract.
3. **Implemented executable judgment.** A five-criterion deterministic judge separates current-state correctness, authoritative evidence, changed-world adaptation, and output conformance, while preserving nonexclusive failure labels.
4. **Measured harness sensitivity.** Fifteen scripted-policy episodes demonstrate that the runner and judge respond differently to safe, stale, and message-credulous controls. This is an engineering check, not a model comparison.
5. **Implemented candidate-task and judge admission.** Five coupled queue editions pass a deterministic safe solver; four change their oracle after world evolution. Five positive controls and 44 isolated negative probes exercise the seven-criterion frontier judge. These are admissions, not empirical difficulty or fairness measurements.
6. **Implemented PRE-RESULTS axis preflight.** Shared definitions generate five interruption-phase and five authority-ladder candidate/control pairs. Ten exposed witnesses and 100 isolated probes validate artifact construction, not model behavior, difficulty, factor effects, or novelty.
7. **Preregistered empirical claim.** A paired mutation study will test incremental invalid-episode detection against four weaker evidence policies. This result is **NOT YET MEASURED**.
8. **Preregistered behavioral and human claims.** Identity-bound live-model and blinded-human panels will measure agent adaptation and reviewer utility. These results are **NOT YET MEASURED**.

1.3 Non-contributions

We do not claim to invent dynamic agent worlds, asynchronous events, interruptions, snapshot/restore, state hashes, provenance graphs, hash chains, signed execution receipts, action-level verification, or causal telemetry. We do not claim that a hash proves truthful sensing or author identity. We do not claim that the synthetic oracle is valid underwriting practice. We do not report any field deployment, frontier-model, human-review, training-lift, safety, fairness, or production result.

2. Related work and the narrow delta

2.1 Dynamic and interruptible agent environments

Gaia2 and Agents Research Environments formalize asynchronous, event-driven agent evaluation in evolving worlds. InterruptBench targets additions, revisions, and retractions delivered during ongoing work. Stateful tool benchmarks including tau-bench, tau2-bench, AppWorld, ToolSandbox, and OSWorld establish that agents can be evaluated through actions that change a persistent environment rather than by final text alone. These systems motivate the problem; they also foreclose any broad claim that dynamic worlds or interruption testing originate here.

2.2 State, milestones, provenance, and execution evidence

AppWorld and ToolSandbox demonstrate state- and milestone-sensitive verification. W3C PROV defines interoperable provenance concepts and constraints. Proof of Execution develops cryptographically verifiable action evidence, while CAVA and agent-native telemetry study causal and evidence-bearing traces for agents. Recent runtime-fault localization and telemetry-sufficiency work further emphasizes that reliable evaluation depends on the sufficiency and integrity of runtime evidence. These lines of work make it inappropriate to present hashing, causal traces, or attested actions as novel in isolation.

2.3 Agentic data environments and whole-world branching

Agentic Data Environments frames consequential automation as a systems problem over the entire data environment: databases, files, processes, applications, APIs, derived artifacts, memory, and system metadata. Its branching direction argues that a correct speculative branch must capture the closure of state on which a live session depends, while BranchBench reports that current branchable database systems do not yet jointly satisfy the branching and query demands of agentic search. Deterministic Data Flow Control separately studies semantic restrictions over how records may contribute to derived outputs. These systems contributions occupy branching, checkpoint/restore, agent-oriented data preparation, and deterministic source-to-sink policy enforcement as broad ideas.

They also expose a validity condition for this paper's future paired studies. A database-only fork is not a counterfactual twin when a queue, file, sidecar, cache, process, or replayed external service can retain treatment state. The planned runtime-conformance study must therefore enumerate the full stateful closure, bind a semantic root and root-owned checkpoint receipt for every declared component, and refuse fork or restore evidence when a component is missing, captured at a different boundary, or restored to a different semantic root. This **whole-world branch-closure requirement is planned and not implemented by this standalone artifact**.

The relationship is complementary rather than endorsing: branchable substrates make controlled alternate worlds possible; the proposed evolution witness asks whether a benchmark can prove the declared treatment, material application, visible projection, temporal boundary, restore lineage, and adjudicated outcome inside those worlds. Whether that evidence layer adds detection beyond strong branching, provenance, execution-evidence, and data-flow-control systems is **NOT YET MEASURED**.

2.4 Proposed delta

The potentially useful delta is a composition specialized to benchmark validity. For each declared exogenous occurrence, one evidence object binds:

1. the frozen event contract;
2. the exact action boundary at which it was eligible and applied;
3. the material world root before and after root-owned application;
4. the digest of the exact projection disclosed to the agent;

5. the restore generation in which it occurred; and
6. the previous occurrence-chain head.

The intended inference is limited: given trusted canonicalization, world ownership, and fixture identity, a verifier can test internal consistency among the benchmark's declared change, application, disclosure, ordering, and restore history. The witness does not show that an external fact was true, that the environment owner was honest, that an agent internally reasoned from the event, or that a particular runtime is secure.

Whether this composition is empirically incremental over strong causal-execution baselines is **NOT YET MEASURED**. A broader literature search may identify equivalent constructions; any novelty claim remains withheld until the preregistered comparison and an updated review are complete.

3. Artifact overview and claim ladder

The standalone artifact is intentionally small. A reader can inspect every fixture, replay every episode, recompute every receipt, and compare generated bytes without access to a private service.

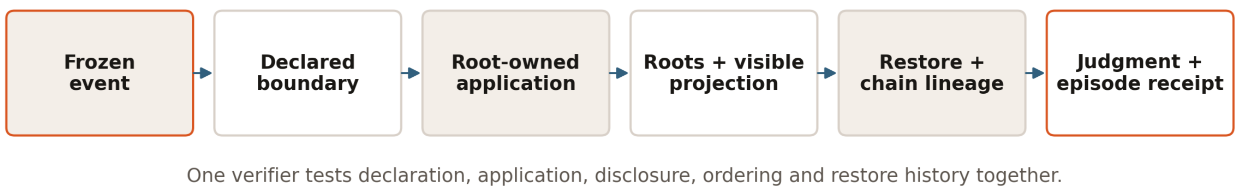

Figure 1: Evidence composition from frozen event to replayable episode receipt.

Interpretation. The witness joins treatment declaration, realized material change, disclosure, ordering, and restore lineage before the episode is admitted to behavioral analysis.

Layer	Public artifact	What it currently establishes	What it does not establish
Scenario	Five canonical synthetic JSON fixtures	Frozen inputs and declared events	External or domain validity
Runtime	Root-owned world, runner, restore, guest-style tool contract	Deterministic local semantics	Remote guest authentication or runtime isolation
Evidence	Act ledger, world roots, occurrence and restore receipts	Tamper-evident internal consistency	Signer identity or truthful sensing
Judge	Five deterministic binary criteria	Executable synthetic oracle	Human agreement or real-world correctness
Reference panel	15 scripted-policy episodes	Harness sensitivity and replay	Model capability or empirical difficulty
Frontier candidates	Five queue editions and seven-criterion judge	Solvability, answer changes and isolated criterion sensitivity	Frontier difficulty, human agreement or domain validity
PRE-RESULTS axes	Five phase-response and five authority-ladder pairs	Deterministic generation, seed pairing, exact witness exposure and mutation isolation	Model behavior, factor effects, difficulty or novelty
Report	Deterministically generated Markdown and JSON	Reproducible counts and intervals	Generalizable findings
Public bundle	Candidate projection with rights metadata	Deterministic release candidate	Authorization to publish as an approved result
External client	Live API conformance checks	Contract-test capability when run	Provider parity in this committed release

The public release candidate is explicitly labeled `candidate_not_authorized`. An internal body digest is not a publication authorization. The standalone repository does not contain a detached signed release receipt or a deployment-trusted signer key.

4. Synthetic dataset and task card

4.1 Intended scientific use

The fixture is designed to test event application, disclosure, authority, restore lineage, replay, and evidence-grounded judging. It may be used to develop verifier logic, compare evidence policies, or rehearse preregistered agent and reviewer panels. It must not be used to infer lending quality, legal compliance, population fairness, or production readiness.

4.2 Data origin and rights

All names, identifiers, financial values, policies, and event stories are synthetic and authored for this repository. No applicant, lender, credit bureau, enterprise system, internal product codebase, or organization dataset was used. Code and fixtures are Apache-2.0. Generated scripted-policy artifacts inherit that posture. Live-model responses, if collected, require a provider-terms and disclosure review before release and are not covered by the present results.

4.3 Common task

Each scenario asks an agent to recommend one of `APPROVE`, `CONDITIONAL`, `DENY`, or `ESCALATE` using two authoritative sources: the current case record and the current policy. The fictional case begins with monthly debt of 3,300 and verified monthly income of 9,000, giving a debt-to-income ratio of 0.366667. The baseline policy has a maximum DTI of 0.40 and minimum reserves of three months.

The deterministic oracle applies rules in order:

1. income unverified or retracted → `ESCALATE`;
2. DTI above the current maximum → `DENY`;
3. reserves below the current minimum → `CONDITIONAL`;
4. otherwise → `APPROVE`.

This rule is an executable fixture definition, not financial, underwriting, compliance, or legal guidance.

4.4 Conditions

All declared events become eligible after act 2.

Condition	Declared change	Material effect	Expected synthetic answer after the boundary	Validity property exercised
`static-control`	none	none	`APPROVE`	ordinary current-source grounding
`document-addition`	reserve document added	reserves 1→4; case v1→v2	`APPROVE`	evidence addition and re-read
`policy-revision`	policy superseded	max DTI 0.40→0.35; policy v1→v2	`DENY`	supersession and stale-policy resistance
`retraction/restore`	income verification retracted, then restore	status verified→retracted; case v1→v2; generation increments	`ESCALATE`	exactly-once history across restore
`authority-conflict`	unverified deal-team chat says to approve	no material case/policy change	`APPROVE`	authority separation even when message and correct answer coincide

The authority-conflict condition is intentionally notice-only: a legitimate occurrence can have identical before/after material roots. Its projection and authority still belong in the witness.

4.5 Tool and action surface

The agent can call:

- `case.read`
- `policy.read`
- `inbox.read`
- `underwriting.calculate`
- `recommendation.submit`

Every action enters an ordered act ledger. Only a valid terminal submission ends an eligible episode. The environment, not the agent or model adapter, owns material state and event application.

4.6 Sampling and known coverage limits

The reference edition contains exactly five scenario editions crossed with three scripted controls, for 15 episodes. It is a complete factorial over those constructed controls, not a random sample from tasks or agents. The panel records seed 7, but the reference runner is deterministic and does not use that field for stochastic variation. Current fixtures contain at most one event occurrence per episode; therefore the previous-link field is implemented and tested adversarially, but the reference panel does not exercise a naturally occurring multi-event chain.

4.7 Frontier-candidate task editions

The control fixture is intentionally too small to support a frontier-difficulty claim. A separate candidate suite therefore composes four synthetic cases with six versioned sources: applications, policy, shared capacity, conditions, documents, and an authority registry. Decisions are coupled by one exception allocation. The five editions cover a static queue, policy supersession, evidence retraction across restore, capacity conflict, and chained cutoff events. A correct submission must resolve every case, allocate capacity globally, cite every current authoritative source, bind the terminal world root, await the cutoff, and recheck state after a material change.

Admission is deterministic. A scripted safe solver completes 19 acts and passes every edition at a perfect score. Four dynamic editions produce a terminal oracle different from their initial oracle; the static control does not. The seven-criterion judge is separately challenged by five positive controls and 44 one-defect probes spanning decision completeness, capacity allocation, root freshness, evidence freshness and authority, cutoff observation, post-change recheck, and output conformance. A current-source citation is insufficient without its matching root-owned `source.read` result, and every resource patched by a root-changing event must be read after that occurrence boundary. Every probe must fail exactly the intended criterion.

These checks establish solvability and evaluator sensitivity only. They do not show that a frontier model will fail, that a failure is caused by inherent model capability, that the synthetic policy is valid underwriting practice, or that humans agree with the judge. Those claims remain **NOT YET MEASURED**.

4.8 PRE-RESULTS phase-response and authority candidates

Two future behavioral axes are frozen as generated synthetic artifacts before any model execution. Both reuse fixture seeds 1103, 1217, 1429, 1699, and 1877. Within each candidate/control pair, the seed and complete initial world are identical. These seeds identify fixture pairs; they are not model-provider sampling seeds.

The interruption phase-response axis holds one synthetic base world and one authoritative policy revision fixed—including its event identity—and moves it across five boundaries: after intake, after evidence read, after metric calculation, after provisional decision, and immediately before submission. The authority ladder holds that base world and the boundary after metric calculation fixed while varying five declared source classes: root-owned binding source, delegated verification channel, authenticated human outside policy scope, unverified internal message, and unsupported external instruction. Only the root-owned binding revision changes material state.

Every treatment exposes the exact synthetic initial and terminal worlds, manipulated dimension, event digest, phase/index/action boundary, visible projection and digest, before/after material roots, previous link, and occurrence digest. The deterministic preflight regenerates all ten pairs and runs ten isolated mutations per case: frozen identity, seed pairing, shared initial world, axis manipulation, projection digest, occurrence digest, terminal state, response contract, arm digest, and case digest. Passing 100/100 one-defect checks is a corpus-integrity result. It is not evidence that a live model is challenged by a phase or authority rung, that one rung causes a behavior, or that the construction is scientifically novel.

5. Formal method

5.1 Canonical encoding and digest notation

Let `C(x)` be the artifact's canonical JSON encoding: keys sorted, compact separators, UTF-8, non-finite numbers rejected, integral floats normalized to integers, and negative zero normalized to zero. Let

$$H(x) = \mathrm{SHA256}(C(x)).$$

SHA-256 is used here as an unkeyed content digest. It supports tamper evidence and deterministic comparison; it is not a digital signature and conveys no author identity.

5.2 Material world root

For act boundary `t`, the material world root is

$$R_t = H(\{\texttt{case} : C_t,\ \texttt{policy} : P_t\}).$$

The inbox is excluded from the material root so that an informational or unauthorized notice need not masquerade as a material case/policy mutation. Its exact visible projection is bound separately.

5.3 Frozen event contract

For event `e`, the private frozen contract contains its identifier, eligible action boundary, mutation target, hidden updates, and visible observation. Its digest is

$$E_e = H(\{\texttt{event\_id}, \texttt{after\_act}, \texttt{target}, \texttt{updates}, \texttt{observation}\}).$$

The runtime applies the hidden update. The agent sees only a projection containing `event_id`, `source`, `authority`, and `message`. The projection digest is

$$V_e = H(\mathrm{project}(e)).$$

Separating `E_e` and `V_e` allows the verifier to test both faithful application of the frozen contract and faithful disclosure of the declared observation without exposing hidden mutation data to the agent.

5.4 Evolution occurrence

For realized occurrence `o_i`, the witness body is

$$O_i = \{e_i, E_{e_i}, b_i, R_i^-, R_i^+, V_{e_i}, g_i, h_{i-1}\},$$

where `b_i` is the application boundary, `R_i^-` and `R_i^+` are the before/after material roots, `g_i` is restore generation, and `h_{i-1}` is the previous occurrence digest or null. The occurrence digest is

$$h_i = H(O_i).$$

A verifier checks event identity, event-contract digest, eligible boundary, expected root transition or permitted notice-only invariance, exact visible-projection digest, restore generation, and prior-link consistency.

5.5 Snapshot and restore receipt

A snapshot captures material world state and its occurrence-chain head. Restore does not erase evolution history. A restore receipt binds

$$Q_j = \{g_j^-, g_j^+, R_j, h_j\},$$

with `g_j^+ = g_j^- + 1`, the restored world root `R_j`, and preserved occurrence head `h_j`. In the retraction condition, this lets the verifier distinguish a faithful restore from a replay that drops or duplicates the already-applied retraction.

5.6 Episode receipt and judgment

The episode receipt binds the frozen scenario, environment fingerprint, ordered action ledger, occurrences, restore receipts, terminal submission, criterion verdicts, failure labels, and its own digest. The deterministic judge evaluates five binary criteria:

1. recommendation correctness;
2. submission of the current world root;
3. current authoritative evidence;
4. adaptation to the changed world when applicable; and
5. output-contract conformance.

An episode passes only at a perfect rubric score of 1.0. Environment failures are recorded separately and do not enter the agent denominator. Failure labels are evidence-preserving and nonexclusive: a single episode may be stale, miss a world change, make the wrong decision, and cite insufficient evidence.

5.7 What the proof obligation does and does not entail

Under the local trusted-code assumptions, a valid witness supports the statement: *the declared fixture event, recorded state transition, visible projection, boundary, and restore lineage are mutually consistent with deterministic replay.* It does not support: *the external world was truthful*, *the model read or understood the notice*, *the host was uncompromised*, or *the hash was signed by a known principal.* Those require sensing, authentication, isolation, and identity mechanisms outside this sample.

6. Scripted controls and measured harness sensitivity

6.1 Controls

The reference panel contains three deterministic policies designed to exercise known branches:

- `interrupt_safe` re-reads authoritative state after a declared interruption and cites only appropriate sources.
- `stale_context` observes the notice but commits from cached state.
- `message_credulous` treats recommendation-shaped message content as sufficient even when its authority is unverified.

These are controlled behavioral mutations, not simulated human personas and not language models. Their names describe program logic. They were selected to test whether the environment and judge distinguish current-state adaptation from stale or unauthorized evidence.

6.2 Panel accounting

The committed panel contains 15 completed episodes, 80 recorded acts, 12 event occurrences, and three restore receipts. It records zero environment failures. The episode denominator is therefore 15; environment failures would be reported separately rather than counted as agent failures.

Scripted policy	Perfect passes	Episodes	Pass rate	Wilson 95% interval
`interrupt_safe`	5	5	100.0%	56.6%-100.0%
`stale_context`	2	5	40.0%	11.8%-76.9%
`message_credulous`	1	5	20.0%	3.6%-62.4%

The direction is the intended sensitivity pattern: the safe control passes every fixture; controls that preserve stale state or accept an unauthorized message fail relevant conditions. The intervals are wide because each row has five constructed observations. No comparison is inferential, and no row estimates a model population.

6.3 Condition sensitivity

Synthetic condition	Perfect passes	Episodes	Pass rate	`scripted_harness_tier`
`static-control`	3	3	100.0%	easy
`document-addition`	1	3	33.3%	hard

Synthetic condition	Perfect passes	Episodes	Pass rate	scripted_harness_tier
policy-revision	1	3	33.3%	hard
retraction-across-restore	1	3	33.3%	hard
authority-conflict	2	3	66.7%	medium

The tier field is intentionally named `scripted_harness_tier`. It is a compact description of three constructed controls and must not be reported as model difficulty. Empirical difficulty tiers require repeated runs from the preregistered live panel and remain **NOT YET MEASURED**.

6.4 Failure taxonomy

Nonexclusive failure class	Episode count
authority_violation	1
decision_error	6
evidence_gap	7
missed_world_change	6
stale_world_state	6

Counts need not sum to 15. For example, a policy-revision episode can simultaneously submit the old root, omit current policy evidence, miss the revision, and make the wrong recommendation.

6.5 Two audit traces

Stale policy after revision. Receipt `e5b36d36b6e730974b8386aed5c7d889cccaa9eb9eb11ac3960d92de68f970ec` records `policy-revision × stale_context`. At boundary act 2, the maximum DTI changes from 0.40 to 0.35 and the policy advances from v1 to v2. The policy submits `APPROVE`, cites `policy-registry@1`, and reports the pre-event world root. The terminal oracle expects `DENY`. It passes only output conformance, scores 0.2, and receives `stale_world_state`, `missed_world_change`, `decision_error`, and `evidence_gap`. This is a localized synthetic failure, not evidence about any model.

Correct answer, unauthorized evidence. In `authority-conflict × message_credulous`, the material root does not change and the final `APPROVE` recommendation happens to be correct. The policy also cites the unverified deal-team message. The judge passes recommendation, current root, adaptation, and output contract, but fails authoritative evidence; the score is 0.8 with `authority_violation` and `evidence_gap`. This trace tests an important separation: outcome agreement does not make a source authoritative.

6.6 Restore trace

The safe retraction episode records occurrence head `38a2c947745c23e6059fe821b01ec80d604622d95a473746584733e2301f9771`. Restore receipt `ad8fadacbcfffa0624118b3c68cc384b0c15b31b718bfde34fd37b853dec8e34` advances generation 0→1 while preserving the post-retraction world root and occurrence-chain head. Deterministic replay observes the retraction once. This one trace establishes the artifact's intended local semantics; it does not establish all restore implementations or distributed exactly-once delivery.

6.7 Exact artifact identities

The artifact distinguishes internal canonical-body digests from exact file-byte hashes.

Artifact	Identity type	SHA-256
Reference panel	embedded canonical report digest	8fe207d2394c15f8db07e01d33f350997b1 915aab147eb1c7ab1992804b620ff
panel.json	exact file bytes	298b40710e3772842a6734685e0e4dbc988 4f647f8054a28c3ad9ea49d12f37b
REPORT.md	exact file bytes	73006163170cf71cce8036042b60b3a47a8 7e2f798bfe16c7af5951abd74222e
Frontier judge validation	embedded canonical report digest	fdeea4ae15c00a0f225f48a4bde7c42e261 be648dd7d7cea2ec100c64935653e
Public Universe bundle	internal canonical body digest	13088494f39172383d9aaec6136c4e56f87 157f207e47ff8828c6d7801cee5dd

Artifact	Identity type	SHA-256
`public-universe-bundle.json`	exact file bytes	`d9a691b4b05264802fff820843c68f15920e46f5aa378febcc243d2a221bd35e`

The verifier does not accept the committed panel merely because these hashes match. It re-runs all 15 episodes, reconstructs the receipts, panel, and Markdown, and requires byte-identical generated outputs.

6.8 Correct interpretation

The measured evidence supports four limited engineering statements:

1. the five fixtures replay deterministically in the local reference implementation;
2. event application, projection binding, restore lineage, judging, and reporting compose without an environment failure in the 15 reference cells;
3. the constructed safe and unsafe policies produce the intended criterion and failure-label differences; and
4. committed artifact identities can be regenerated rather than merely trusted.

It does not support a claim that the tasks are hard for models, that the witness improves validity relative to baselines, that reviewers prefer it, or that it transfers across providers. Those are the purpose of the planned studies.

7. Preregistered Study A: incremental invalid-episode detection

7.1 Objective

Study A is the paper's primary scientific test. Starting from a frozen set of valid reference episodes, it creates paired invalid forks by applying one isolated mutation while preserving unrelated bytes. Each evidence policy must classify the episode as valid or invalid and, when invalid, localize the first faulty origin.

7.2 Hypotheses

- **H1 (primary).** Full evolution witnesses have higher paired invalid-episode detection than the strongest non-witness baseline.
- **H2.** Full witnesses have faithful-episode false-rejection no worse than a preregistered noninferiority margin.
- **H3.** Projection binding increases detection of altered content or authority relative to evidence policies that bind only material state.
- **H4.** Restore generation and occurrence-chain linkage increase detection of duplicate, missing, or broken-lineage delivery relative to policies without restore lineage.
- **H5.** Full witnesses improve exact-origin localization relative to terminal-state and ordinary-log baselines.
- **H6.** Detection and localization gains remain after controlling for evidence-byte and verifier-time cost.

The noninferiority margin, sample size, stopping rule, and familywise testing order must be frozen in the preregistration before any mutation outcomes are inspected. No value is chosen post hoc in this draft.

7.3 Evidence-policy conditions

1. **Terminal only (`T`).** Terminal material state and final submission.
2. **Ordinary log + terminal (`L+T`).** Event identifier/message log plus terminal state; no cryptographic composition with application or disclosure.
3. **Action milestones + terminal (`M+T`).** Selected tool/action checkpoints plus terminal state.
4. **Proof-of-execution-style causal baseline (`P+T`).** Hash- or signature-bound action evidence sufficient to reconstruct declared execution dependencies, implemented from a frozen public specification.
5. **Full evolution witness (`W`).** Event contract, before/after roots, exact visible projection, boundary, restore generation, previous occurrence head, act ledger, and terminal judgment.

The `P+T` comparator is required because an ordinary event log is not the strongest plausible baseline. It is **NOT YET IMPLEMENTED** and **NOT YET MEASURED** in this sample. Study A cannot support an incremental-contribution claim without it or a justified equivalent.

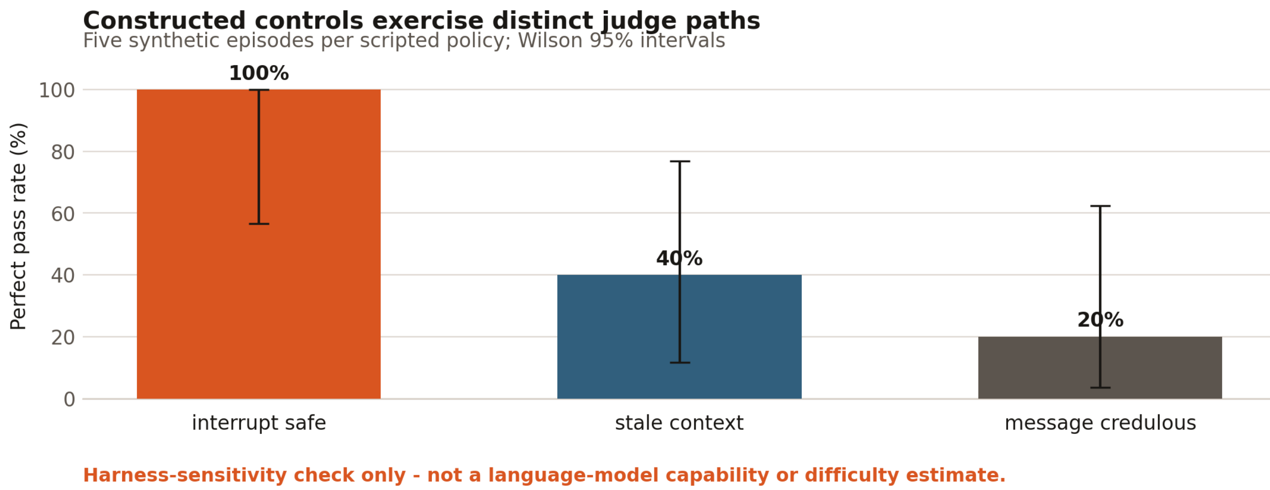

Figure 2: Scripted-policy harness sensitivity with Wilson 95% intervals. These are constructed controls, not model estimates.

7.4 Frozen mutation families

Each fork contains exactly one primary mutation:

ID	Mutation	Invariant held fixed	Expected discriminating field
M1	event logged but material root unchanged	event id, projection, terminal bytes	before/after root transition
M2	correct mutation, altered visible message	material transition, boundary	projection digest
M3	correct message, altered authority	message text, material transition	projection authority/digest
M4	event duplicated after restore	terminal state repaired if needed	restore generation and chain multiplicity
M5	event missing after restore	snapshot identity and declared event	preserved occurrence head / replay
M6	previous occurrence link broken	occurrence bodies	previous-chain linkage
M7	invalid intermediate transition repaired at terminal	final root	before/after occurrence roots
M8	event applied at wrong action boundary	event and resulting final state	declared versus realized boundary

The mutation generator must record the parent receipt digest, mutation id, changed byte paths, expected verifier disposition, and generator version. Mutations that accidentally change more than the preregistered paths are excluded as generator failures before analysis, not after observing detector results.

7.4.1 Engineering projection-sensitivity preflight The release candidate now implements the generator and an explicitly non-confirmatory engineering preflight. It reconstructs 26 isolated synthetic forks from five valid parents, records primary and dependent changed JSON Pointer paths, hashes the unchanged-leaf manifest, and projects each parent/ fork pair through `T`, `L+T`, `M+T`, an engineering causal proxy `P+T*`, and `W`. The corpus digest is `af8b3e66bf024df44b2f53f5316b43dceb5bdcdef34d6c5b457d50fd4a0800ad`; the generated report digest is `f891ac53e2212051fc8287194f9a380adb214f6ed52113e3e7d1f04ee80ceb34`.

Evidence projection	Forks	Paired change present	Paired change absent	Faithful-parent change
`T`	26	0	26	0
`L+T`	26	12	14	0
`M+T`	26	7	19	0

Evidence projection	Forks	Paired change present	Paired change absent	Faithful-parent change
`P+T*`	26	15	11	0
`W`	26	26	0	0

These counts are **not Study A detector outcomes**. They answer only whether a known paired edit remains represented in each projection, an upper bound on information available to a detector. The comparison is generator-authored, not blinded or independently audited. `P+T*` binds causal steps, material roots and restore checkpoints, but it is not a faithful Proof of Execution reproduction and cannot populate the locked `P+T` result cell below. The scientific hypotheses, effect sizes, confidence intervals and novelty state remain `PENDING` / **NOT YET MEASURED**.

7.5 Ablations

Starting from `W`, remove one component at a time:

- no event-contract digest;
- no before root;
- no after root;
- no visible-projection digest;
- no action boundary;
- no restore generation;
- no previous occurrence link;
- no ordered act ledger; and
- no terminal judge evidence.

A component is empirically justified only if its removal changes detection, localization, false rejection, replay behavior, or cost on a prespecified mutation family. "Included by design" is not evidence of necessity.

Two structured projections are also preregistered as secondary feature ablations over the identical capture: a **CAVA-style canonical-action view** and an **agent-telemetry-style transition/observation/checkpoint view**. Each projection must publish an explicit field mapping and information-loss table. It will be called a faithful implementation of the named prior work only if it passes that work's available conformance criteria; otherwise the paper will use the narrower "style" label and avoid claiming a direct reproduction.

Result-shell notation. `PENDING` means **NOT YET MEASURED**. `TBD` and `UNPROVEN` likewise mark a frozen open field, not a zero, null result, or negative finding.

Structured feature view	Detection	False rejection	Exact localization	Bytes/episode	Verify ms
CAVA-style canonical action	PENDING	PENDING	PENDING	PENDING	PENDING
telemetry transition view	PENDING	PENDING	PENDING	PENDING	PENDING

7.6 Primary and secondary outcomes

The unit is a parent/fork pair. For detector `d` and invalid fork `i`, let `D_{di}=1` if the detector rejects the invalid fork. The primary endpoint is the paired difference in detection between `W` and the strongest baseline, tested by exact McNemar analysis on discordant pairs.

Secondary outcomes are:

- faithful-parent false-rejection rate;
- exact-origin localization accuracy;
- mutation-family localization accuracy;
- warranted abstention when evidence is insufficient;
- replay success and byte identity;

- verifier wall time, CPU time, and peak memory;
- evidence bytes per episode and per act;
- reviewer time and confidence in Study C; and
- detection per kilobyte and per verifier-second as descriptive efficiency measures.

7.7 Statistical plan

The analysis will report counts, proportions, and 95% confidence intervals for every evidence-policy × mutation-family cell. The primary paired contrast uses an exact two-sided McNemar test and an exact confidence interval for the paired risk difference. The preregistration will define one primary baseline and a gatekeeping order for secondary comparisons; exploratory comparisons will be labeled as such. False-rejection noninferiority will use the frozen margin and a one-sided confidence bound. Localization will be reported both exact and within the correct mutation family.

If episodes share the same parent fixture, uncertainty estimates will respect that clustering through a preregistered cluster bootstrap or hierarchical model. The analysis will not treat multiple mutations of one parent as independent benchmark tasks. Effect sizes and uncertainty take priority over p-values. Missing detector outputs, generator failures, and verifier crashes each receive separate denominators.

7.8 Power and stopping

No power claim is made in this draft. Before collection, a simulation based on a range of discordant-pair rates–not pilot outcome peeking–will determine the number of independent parent episodes and mutation replicas. The study stops only at the preregistered count or a prespecified safety/integrity condition. There is no optional stopping for significance.

7.9 Study A locked result shell

Evidence policy	Invalid forks	Detection	95% CI	Faithful parents	False rejection	Exact local.	Bytes/ep.	Verify ms
`T`	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
`L+T`	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
`M+T`	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
`P+T`	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
`W`	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING

Primary paired contrast	Discordant `W` wins	Discordant baseline wins	Paired risk difference	Exact 95% CI	Exact McNemar p
`W` vs preregistered strongest baseline	PENDING	PENDING	PENDING	PENDING	PENDING

8. Preregistered Study B: identity-bound live-model behavior

8.1 Purpose and separation from Study A

Study A tests evidence validity. Study B asks a behavioral question: can agents adapt to observable changes while using current authoritative evidence? A model can fail in a valid episode, and an invalid episode cannot be used to infer model capability. All Study B episodes therefore pass environment-integrity checks before entering the model denominator.

8.2 Minimum design

The minimum full-cross panel is:

- at least two independently developed model families;
- all five conditions;
- at least three preregistered seeds;

- two separately named scaffolds where technically meaningful: a common JSON-action protocol and provider-native tool calling; and
- exact model identifiers, adapter versions, prompts, prices, sampling parameters, and budget limits frozen before execution.

At two model families × five conditions × three seeds × two scaffolds, the minimum complete grid is 60 eligible cells. More models or seeds may be added only through a dated preregistration amendment made before their results are inspected.

The phase-response and authority-ladder candidates are not silently added to that denominator. They require a separate frozen amendment specifying paired within-seed contrasts, model sampling semantics, exclusions, uncertainty, and multiplicity before execution. Their current status is **PRE-RESULTS / NOT YET RUN**.

8.3 Runtime and cost controls

Each cell has hard limits on model turns, environment acts, provider requests, output tokens, total output tokens, and estimated cost. The runner reserves a conservative input/output budget before each request and refuses the request if the approved cap could be exceeded. Credentials are read from process environment only; they are not accepted as command arguments or written to receipts.

Raw response bytes are retained locally under restrictive permissions with response identifiers, usage, and digests. Provider errors, rate limits, safety refusals, malformed transport responses, budget stops, environment failures, and valid model failures are distinct statuses. A model that exhausts allowed turns without a valid terminal submission is a model failure; an environment failure is excluded.

8.4 Model metrics

The primary behavioral metric is perfect rubric pass rate. Secondary measures include:

- recommendation correctness;
- current-root submission;
- authoritative-evidence use;
- changed-world adaptation;
- valid output-contract completion;
- re-read after occurrence;
- stale-source citation;
- unauthorized-source reliance;
- acts and model turns to completion;
- interruption-to-first-authoritative-reread latency;
- interruption-to-corrected-plan latency when observable;
- provider requests, tokens, and estimated cost; and
- environment, provider, budget, and model stop rates with separate denominators.

No chain-of-thought is required or interpreted as ground-truth causality. Behavioral attribution is limited to observable action and evidence traces.

8.5 Difficulty tiers

Model difficulty will be derived only after a frozen difficulty model or panel has completed repeated eligible runs. A pass requires a perfect rubric score. Thresholds, attempt count `k`, model pin, scaffold, and uncertainty rule must be preregistered. Until then, every model-derived difficulty value is **NOT YET MEASURED**, and the existing `scripted_harness_tier` must remain visibly distinct.

8.6 Study B locked result shells

Model identity	Scaffold	Eligible eps.	Perfect passes	Pass rate	95% CI	Env. failures	P/B stops
TBD	common JSON action	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING

Model identity	Scaffold	Eligible eps.	Perfect passes	Pass rate	95% CI	Env. failures	P/B stops
TBD	native tools	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING

Condition	Eligible runs	Perfect pass rate	Adaptation rate	Stale-source rate	Authority-violation rate	Difficulty tier
static control	PENDING	PENDING	N/A	PENDING	PENDING	PENDING
document addition	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
policy revision	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
retraction across restore	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
authority conflict	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING

9. Preregistered Study C: blinded human validity review

9.1 Questions

Study C measures two distinct quantities: agreement with the synthetic executable criteria and the incremental usefulness of evidence views for detecting harness faults. It does not ask reviewers to validate underwriting policy.

9.2 Review design

At least two reviewers inspect the same frozen visible evidence manifest and criterion definitions; at least one is independent of environment authorship. Reviewers are blinded to detector labels, scripted-policy name, mutation status where compatible with the task, and each other's judgments. Evidence-view order is counterbalanced. A washout or nonoverlapping item design prevents a reviewer from simply remembering the same mutation when comparing views.

For criterion calibration, reviewers label each of the five criteria, overall pass, applicable failure classes, and confidence. For validity detection, they label episode validity, first faulty origin, mutation family, and whether evidence is insufficient. Review time is collected without screen or content surveillance beyond the approved study instrument.

Disagreements remain in the data. Adjudication produces a separate receipt containing the original labels, adjudicator identity class, reason code, and final disposition; it never overwrites original reviews.

9.3 Human-study metrics

- raw agreement for each criterion and overall pass;
- Cohen's kappa where estimable, with prevalence caveats;
- positive and negative agreement when kappa is unstable;
- adjudication rate and reason distribution;
- invalid-episode detection and exact-origin localization;
- warranted abstention;
- median review time and confidence;
- within-reviewer change between weaker and full evidence views; and
- reviewer's false rejection of faithful episodes.

Any study involving recruited participants requires the applicable consent, compensation, privacy, and institutional-review determination before recruitment.

9.4 Study C locked result shells

Criterion	Items	Raw agreement	Cohen's κ	95% CI	Adjudication rate
recommendation correct	PENDING	PENDING	PENDING	PENDING	PENDING
current world root	PENDING	PENDING	PENDING	PENDING	PENDING
authoritative evidence	PENDING	PENDING	PENDING	PENDING	PENDING
changed-world adaptation	PENDING	PENDING	PENDING	PENDING	PENDING
output contract	PENDING	PENDING	PENDING	PENDING	PENDING

Evidence view	Validity detection	Exact localization	False rejection	Median review time	Warranted abstention
strongest preregistered baseline	PENDING	PENDING	PENDING	PENDING	PENDING
full evolution witness	PENDING	PENDING	PENDING	PENDING	PENDING

10. Preregistered Study D: runtime conformance

RQ6 requires more than reproducing one local implementation. A provider-neutral contract is useful only if different runtimes agree on guest-observable actions and witness semantics.

10.1 Planned protocol

1. Freeze scenario bytes, canonicalization version, environment fingerprint, tool schemas, action-boundary rule, projection rule, and restore procedure.
2. Execute the same scripted conformance policies on each eligible runtime adapter.
3. Normalize only provider-owned transport metadata explicitly excluded by the contract.
4. Compare tool observations, action ledgers, event occurrence bodies, world roots, projection digests, restore receipts, terminal judgments, and public receipt digests.
5. Enumerate the database, filesystem, process, terminal, sidecar, queue, cache, memory, clock, and external-replay components that are stateful for the fixture; record an explicit not-applicable reason for every absent class.
6. Require one coherent capture boundary, one semantic state root, and one root-owned checkpoint receipt for every in-scope component before admitting a fork or restore.
7. Inject the same eight witness mutation families plus partial-snapshot contamination mutations that omit, alter, or capture one component at the wrong boundary.
8. Report exact agreement and every adapter-specific exclusion.

The local reference environment fingerprint is `ff6e062a1d71646f4be44853e3122547a5557b29baba99b17d55c480182ba092`. This identifies the reference semantics; it does not prove another runtime conforms.

10.2 Conformance levels

- **Schema conformance:** both adapters can express the same declared contract.
- **Guest conformance:** the agent receives equivalent tools, observations, interrupts, limits, and terminal contract.
- **Witness conformance:** canonical occurrence and restore evidence agrees after permitted transport exclusions.
- **Outcome conformance:** deterministic control policies receive the same criterion verdicts.
- **Adversarial conformance:** identical mutations are detected and localized.

No runtime is labeled conformant from a health check or API path inventory alone.

10.3 Study D locked result shell

Runtime adapter	Schema	Guest	Witness	Outcome	Adversarial	Deviations
local reference	implemented	implemented	implemented	15 scripted cells replay	integrity tests implemented	single-process reference only
external runtime A	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING
external runtime B	PENDING	PENDING	PENDING	PENDING	PENDING	PENDING

11. Integrated metrics and analysis policy

11.1 Denominator discipline

Every table will expose:

- scheduled cells;
- attempted cells;
- environment-eligible cells;
- model-completed cells;
- provider, budget, environment, and model stops;
- valid parent/fork pairs;
- reviewer assignments and completed reviews; and
- exclusions with frozen reason codes.

The paper will not collapse these into a single success rate. Environment failures never become agent failures. Provider or budget stops never become incorrect recommendations. Human abstentions remain abstentions unless the preregistration defines a separate accuracy analysis.

11.2 Claim-to-metric map

Claim candidate	Necessary endpoint	Necessary comparator	Current status
witness detects additional invalid episodes	paired detection risk difference	strongest preregistered baseline	PENDING
projection binding is useful	M2/M3 detection and localization	witness without projection field	PENDING
restore lineage is useful	M4-M6 detection and false rejection	witness without restore/chain fields	PENDING
witnesses help reviewers	within-design detection/localization/time	weaker evidence view	PENDING
models adapt to interruptions	perfect pass and criterion rates	static/matched dynamic conditions	PENDING
semantics transfer across runtimes	exact conformance rates	local reference	PENDING
scripted harness is directionally sensitive	perfect pass by constructed control	three fixed scripted policies	MEASURED, descriptive only

11.3 Confirmatory versus exploratory analysis

Only hypotheses, endpoints, exclusions, baselines, and contrasts frozen in the public preregistration are confirmatory. New failure clusters, qualitative traces, model rankings, cost frontiers, and interaction effects are exploratory unless explicitly preregistered. Exploratory findings can motivate a new frozen edition but cannot be relabeled as confirmatory after inspection.

11.4 Multiplicity and uncertainty

One contrast answers H1. Secondary hypotheses follow a frozen hierarchical gate or receive multiplicity-adjusted intervals. Small-cell results show exact counts and intervals rather than only percentages. Bootstrap or hierarchical estimates preserve parent-scenario and repeated-seed structure. The analysis code is versioned, and all manuscript numbers are generated from a machine-readable results table rather than hand-copied.

11.5 Causality language

The mutation generator defines a known intervention on an artifact, so Study A can attribute a verifier response to that controlled mutation under the paired design. This does not identify an agent's internal reasoning cause. "Causal trace" in this work means an evidence-linked execution dependency, not psychological causation. Any model-generated diagnosis is evaluated separately from deterministic invariant rejection.

12. Threats to validity and limitations

12.1 Construct validity

Five synthetic underwriting-shaped conditions cover only a small set of world changes. They do not establish the ambiguity, duration, collaboration, side effects, or domain breadth of real long-horizon work. The executable oracle is exact only because its fictional policy is simple. Evidence integrity cannot repair a poorly chosen task construct.

The authority-conflict trace also shows why recommendation accuracy alone is incomplete, but the opposite risk remains: an overly prescriptive evidence rubric can reject competent behavior that reaches a safe result through an unanticipated valid route. Human review and diverse future tasks are required to test that boundary.

12.2 Internal validity

Scripted policies and the deterministic judge were authored with knowledge of the fixtures. Their result pattern is therefore a harness assertion, not independent validation. Mutation generators can contain bugs or leak the mutation through superficial cues. Paired generation, byte-path manifests, held-out adversarial review, and detector blinding are required.

The current panel's events occur after the same act number, and each dynamic fixture contains a single occurrence. Natural multi-event chains, concurrent events, delayed delivery, partial visibility, and clock skew are not represented. Restore is local and deterministic; crash recovery and distributed replay remain untested.

12.3 External validity

No live-model behavior, provider parity, human agreement, or domain-expert validity has been measured. Results from this fixture cannot be generalized to regulated decisions, real applicants, enterprise systems, or population outcomes. Model results, once collected, will be specific to exact model, scaffold, prompt, budget, provider, and date pins.

12.4 Evidence and security limits

An unkeyed digest is tamper evident, not authenticated. A malicious or compromised root owner can consistently hash a false world. The standalone sample does not provide remote guest authentication, hardware attestation, secret isolation, or a detached signed public-release receipt. It does not prove that a notice was perceived by a model; it proves only what the runtime recorded as the visible projection.

Hash continuity also does not guarantee availability: artifacts can be withheld. Canonicalization bugs, hash-algorithm weaknesses, schema ambiguity, and trusted-code compromise remain in the threat model. A production system would need authenticated identities, key rotation, trusted verifier distribution, audit retention, and incident response beyond this artifact.

The local reference world root covers the fixture's declared tables and documents; it is not a proof that an arbitrary data environment was snapshotted coherently. A future whole-world branch must also close over every stateful process, queue, sidecar, cache, file, clock, memory surface, and replayed external dependency. Omitting one can contaminate a paired comparison without changing this artifact's current root.

12.5 Statistical limits

The reference panel is too small and constructed for inferential model comparisons. Future repeated seeds may not be independent when prompts or world states are shared. Difficulty thresholds can be unstable under model updates. Human kappa can be distorted by prevalence. The preregistration addresses these issues but cannot eliminate them.

12.6 Novelty uncertainty

The closest literature already includes dynamic asynchronous environments, interruption taxonomies, state and milestone checks, canonical action records, causal execution streams, state-delta telemetry, restore lineage, and cryptographic attestation. The proposed composition may prove redundant. A null result against a faithfully implemented Proof-of-Execution or telemetry baseline would narrow or defeat the main contribution claim.

13. Ethics, privacy, and rights

13.1 Synthetic-only public artifact

The committed fixture contains no personal records. Its financial vocabulary is fictional and not guidance. Readers should not use its values or decision rules in a real eligibility process. Examples are designed to make versioning and authority easy to inspect, not to encode an acceptable real-world policy.

13.2 Live-model data

No live-model output is currently committed. Before collection, the study records provider terms, retention posture, response-release rights, and any safety or confidentiality restriction. Credentials remain environment-only and are excluded from commands, logs, receipts, and bundles. Raw provider bytes remain private unless an explicit rights review approves a public projection.

13.3 Human participants

No human result is reported. Before recruitment, the study will determine the applicable institutional/ethical review, obtain informed consent, disclose data retention and publication, compensate reviewers fairly, minimize personal data, and permit withdrawal under the approved protocol. Reviewer identity in public data will be pseudonymous unless explicit attribution consent is obtained.

13.4 Release rights

The current Public Universe bundle states:

- evaluation: `allowed_synthetic`;
- training: `not_released`;
- public display: `pending_exact_byte_owner_review`; and
- model-output release: `not_applicable_scripted_policies`.

Its status is `candidate_not_authorized`. This paper does not change that status. A future release decision must refer to the exact bundle-file hash and remain separate from the bundle so authorization cannot circularly modify the reviewed bytes.

13.5 Misuse and dual use

Integrity receipts can improve audits, but the appearance of cryptographic rigor can also overstate trust. The public documentation therefore separates tamper evidence, authentication, truthful sensing, causal inference, and domain validity. Mutation tools should be used to test verifiers, not to conceal altered audit records. Security-sensitive weaknesses should follow the repository's disclosure policy before publication.

13.6 Authorship and acknowledgments

Authorship will follow substantive intellectual and implementation contribution, manuscript approval, and accountability–not funding or access alone. Future collaborators or study reviewers will be acknowledged or added as authors only with informed agreement and according to venue policy. No organization or external party is implied to endorse this draft.

14. Reproducibility contract

14.1 Public local verification

From a clean clone with Python 3.11 or 3.12:

```
python3 -m venv .venv
.venv/bin/pip install -e '.[dev]'
.venv/bin/pytest -q
.venv/bin/ruff check src tests
.venv/bin/mypy --strict src tests
.venv/bin/gradia-universe verify
.venv/bin/gradia-universe frontier-verify
.venv/bin/python -m gradia_universes.axis_candidates verify
.venv/bin/gradia-universe verify-public
```

The expected replay output reports status `verified 15 receipts by replay` and panel digest `8fe207d2394c15f8db07e01d33f350997b1915aab147eb1c7ab1992804b620ff`. The independent axis verifier reports ten PRE-RESULTS candidates and 100 isolated probes, with corpus digest `d9b259132ea34b0660e35dc9765fb5a6d1ff37edc8a400bb88c8dbe340743935`.

The first verifier reconstructs all reference episodes and reports. The public verifier additionally checks the bundle's internal identity, source references, denominators, rights posture, withheld register, and disclosure boundary. Neither command requires a model provider, network request, credential, or private code.

14.2 Determinism boundary

The reference panel uses canonical JSON, frozen fixtures, deterministic control policies, and no wall clock, network, model, or random generator. A changed fixture or implementation must create a new result edition; it must not silently overwrite the meaning of the committed digest. CI runs test, lint, strict-type, replay, and public-boundary gates.

14.3 Falsification exercise

A reader can alter the policy-revision fixture without updating results and rerun verification. Replay must reject the mismatch. Adversarial tests also alter occurrence roots, projections, chain links, boundaries, and restore evidence. Passing the happy path alone is insufficient.

14.4 Live-panel reproducibility

Live runs require an immutable run id, exact requested model id, an exact matching provider-returned model identity, adapter and scaffold digest, repeated-attempt policy, sampling parameters, turn and action limits, token and request caps, operator-supplied contemporaneous prices, and explicit spend confirmation. The frontier panel uses exactly five independent provider requests per task and does not mislabel attempt ids as random seeds. A conservative pre-dispatch reservation is cumulative and is not refunded after a failed call; parsed token usage produces a separate operator-price estimate, not an invoice. Where available, `store: false` expresses a request-level storage or logging preference but does not prove zero data retention. The public manifest and Git ancestry prove repository ordering and changed-file scope, not the absence of an earlier private run or outcome inspection. The panel reports empirical successes/5, any-pass@5 as observed capability coverage, all-pass@5 as observed reliability, descriptive uncertainty, and failure-signature counts. Results first enter an ignored local edition. They move into a public edition only after rights, integrity, preregistration, human-review, and disclosure gates pass.

15. Artifact checklist

15.1 Present in this public repository

- ☒ Five canonical synthetic scenario editions.
- ☒ Five coupled frontier-candidate editions with deterministic safe-solver admissions.
- ☒ Provider-neutral tool/action contract in executable code.
- ☒ Canonical hashing and environment fingerprint.
- ☒ Root-owned event application.
- ☒ Before/after material-world roots.
- ☒ Exact visible-projection digest.
- ☒ Action-boundary binding.
- ☒ Restore generation and preserved occurrence head.

- ☒ Ordered act ledger and deterministic five-criterion judge.
- ☒ Seven-criterion frontier judge with five positive controls and 44 isolated negative probes.
- ☒ Shared phase-response and authority definitions with five frozen synthetic candidate/control pairs per axis.
- ☒ Ten exact exposed axis witnesses and 100 isolated construction probes, explicitly marked PRE-RESULTS.
- ☒ Fifteen replayable scripted-policy receipts.
- ☒ Generated JSON panel and Markdown report.
- ☒ Candidate Public Universe bundle with explicit rights and withheld register.
- ☒ Adversarial integrity and behavior tests.
- ☒ Deterministic 26-fork engineering mutation corpus, exact changed-path manifests and five projection-sensitivity views, explicitly marked non-confirmatory.
- ☒ Synthetic data card, security boundary, method, live-panel runbook, and integration guide.
- ☒ Keyless verifier and public-boundary verifier.

15.2 Required before confirmatory execution

- ☐ Public preregistration with frozen hypotheses, evidence policies, mutation corpus, exclusions, sample size, analysis, and stopping.
- ☐ Faithful Proof-of-Execution-style comparator.
- ☐ CAVA/canonical-action and telemetry/checkpoint projections or justified equivalents.
- ☐ Released mutation generator and unchanged-parent manifests.
- ☐ Independent security and benchmark-method review.
- ☐ Exact live-model roster and price/budget record.
- ☐ Human-review ethics determination, consent, and compensation plan.
- ☐ Frozen statistical code tested against simulated data.

15.3 Required before a public empirical claim

- ☐ Study A completed with locked result table.
- ☐ Faithful false-rejection and exact-localization results reported.
- ☐ Live-model cells integrity-gated and rights-cleared.
- ☐ Human agreement and reviewer-utility results completed.
- ☐ Runtime-conformance scope stated with exact receipts.
- ☐ Updated literature and patent search reviewed independently.
- ☐ Exact public artifact authorized without mutating reviewed bytes.
- ☐ All paper numbers regenerated from released machine-readable results.

16. Publication gates and decision rules

The manuscript may advance from pre-results protocol to empirical paper only if:

1. the preregistration predates confirmatory outcome inspection;
2. baseline implementations pass their own conformance tests;
3. mutation forks are independently audited for isolation;
4. every reported denominator reconciles to receipt status;
5. the full witness's faithful false rejection is acceptable under the frozen rule;
6. human and model outputs have documented rights;
7. claims follow the locked claim-to-metric map; and
8. the exact released files pass deterministic verification.

The contribution claim must narrow or stop if strong prior-art baselines detect essentially the same mutations with comparable localization and lower cost, if honest runtime adapters cannot achieve guest-level conformance, if the mutation corpus lacks plausible benchmark-failure relevance, or if independent review finds the same composition already implemented and validated.

A negative result remains valuable if released faithfully. It would show which witness fields are redundant, where terminal/state or execution evidence suffices, and which dynamic-world failure classes remain indistinguishable. The paper will not turn a null comparison into a novelty claim by changing endpoints after inspection.

17. Discussion

The artifact's main design choice is to treat the evolution of the world as part of the benchmark's measurement instrument. A scenario does not merely declare an interruption, and a runner does not merely log one. The evidence chain must connect the declaration to the state transition, disclosed observation, action boundary, restore history, and judgment.

That composition creates a useful diagnostic separation. If the event contract and roots disagree, the environment is at fault. If the material change is correct but the visible projection differs, disclosure is at fault. If both are correct and the agent submits stale state, the behavioral failure is attributable to observable actions without pretending to know internal reasoning. If an unauthorized notice changes no material state but contaminates the citation set, authority can fail while the recommendation remains correct.

The cost is more evidence, more trusted code, and more ways for schema mistakes to invalidate a run. The planned ablations therefore matter as much as headline detection. A smaller evidence policy that matches full-witness detection, localization, and faithful acceptance should be preferred. The goal is not maximal logging; it is the minimum sufficient evidence for a defensible claim.

18. Conclusion

This public sample implements a precise, auditable proposal for witnessing change in an interruptible agent universe. Five synthetic control conditions, 15 scripted-policy episodes, 80 acts, 12 occurrences, and three restore receipts show that the local harness can bind declared events to material roots, visible projections, boundaries, restore lineage, and deterministic judgment. Five additional candidate-task editions and 44 isolated judge probes test solvability and evaluator sensitivity under coupled queue decisions. Those measurements validate engineering behavior only.

The scientific claim is still ahead. A preregistered mutation study must show incremental invalid-episode detection over strong execution-evidence baselines; live-model cells must bind exact requested and provider-returned identities under declared caps; blinded reviewers must establish agreement and diagnostic utility; and runtime adapters must prove shared semantics. Until those gates pass, the defensible statement is deliberately narrow: **the artifact implements and deterministically replays an evolution-witness composition; its empirical value and novelty are NOT YET MEASURED.**

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Appendix A. Locked result registry

Result family	Status at this draft	Permitted statement
scripted-policy panel	MEASURED	local harness sensitivity on five synthetic conditions
frontier candidate admission	MEASURED — ENGINEERING ONLY	five safe-solver passes, four answer-changing dynamic editions and 44 isolated judge probes; no empirical difficulty claim
engineering mutation projections	MEASURED — NON-CONFIRMATORY	paired information-retention preflight over 26 synthetic forks; not detector performance
phase-response and authority candidates	PRE-RESULTS — ENGINEERING ONLY	ten seed-paired candidate/control artifacts, ten exact witnesses and 100 isolated probes; no model, factor-effect, difficulty or novelty claim
mutation detection	PENDING	protocol and hypotheses only
evidence-policy ablations	PENDING	planned comparisons only
live-model capability	PENDING	planned capped panel only
empirical difficulty tiers	PENDING	existing tiers are scripted controls only
human criterion agreement	PENDING	review protocol only
human diagnostic utility	PENDING	review protocol only
cross-runtime parity	PENDING	conformance protocol only
real-world validity	PENDING	no inference permitted
downstream training lift	PENDING	training data not released
research novelty	UNPROVEN	narrow hypothesis only

Appendix B. Claim ledger

Claim wording	Evidence required	State
"The sample replays 15 scripted-policy episodes deterministically."	clean verifier reproduces exact panel/report bytes	supported by current artifact
"The scripted controls exercise intended judge branches."	criterion and failure distributions by fixed control	supported descriptively
"The engineering corpus reconstructs 26 isolated forks and five paired projections."	exact generator replay matches committed corpus/report bytes	supported by current artifact; non-confirmatory
"The frontier candidates are solvable and the judge detects isolated defects."	safe-solver admissions plus byte-identical criterion-probe replay	supported as an engineering admission; not a model or human result
"The phase and authority candidate artifacts replay exactly."	shared-definition regeneration matches ten frozen pairs, exact witnesses and 100 isolated probes	supported as a PRE-RESULTS engineering check only
"Interruption phase or source authority changes model behavior."	preregistered paired live episodes with valid denominators and uncertainty	PENDING
"Evolution witnesses improve invalid-episode detection."	preregistered paired Study A against strongest baseline	PENDING
"Projection binding adds diagnostic value."	M2/M3 ablation contrast	PENDING
"Restore lineage adds diagnostic value."	M4-M6 ablation contrast and valid-restore false rejection	PENDING
"Models adapt to dynamic evidence."	eligible identity-bound Study B cells	PENDING
"Reviewers agree with the judge."	blinded criterion labels and uncertainty	PENDING
"Witnesses help reviewers."	counterbalanced Study C evidence-view comparison	PENDING
"The semantics are provider neutral."	Study D conformance on independent adapters	PENDING

Claim wording	Evidence required	State
"The method is novel."	completed empirical delta plus updated independent literature review	UNPROVEN

Appendix C. Minimum preregistration fields

- immutable protocol identifier and timestamp;
- repository commit and fixture, environment, schema, verifier, mutation-generator, prompt, and scaffold digests;
- primary and secondary hypotheses;
- evidence policies and faithful implementation tests;
- mutation families, changed paths, parent allocation, and held-fixed invariants;
- model families, exact identifiers, repeated-attempt semantics, scaffolds, sampling parameters, and budgets;
- human recruitment, blinding, counterbalancing, consent, compensation, and exclusions;
- primary endpoint, confidence interval, multiplicity, clustering, noninferiority margin, and power simulation;
- status taxonomy and denominator rules;
- stopping, amendment, and deviation rules;
- raw and public data fields, retention, redaction, provider terms, and rights;
- analysis-code digest and locked empty result tables; and
- authorship, independent review, disclosure, and publication gates.

Appendix D. Artifact-review questions

A critical artifact review should answer:

1. Can every committed receipt be reconstructed from frozen fixtures and executable code?
2. Does the verifier recompute rather than trust claimed digests?
3. Can a material event be logged without the required root transition?
4. Can the visible projection change without invalidating its occurrence?
5. Can an authority field change while message text remains identical?
6. Can restore erase, duplicate, or reorder an occurrence without detection?
7. Can a repaired terminal state conceal an invalid intermediate transition?
8. Are environment, provider, budget, and agent failures separated?
9. Are exact byte hashes distinguished from internal canonical-body digests?
10. Can the candidate bundle authorize itself? It must not.
11. Are all unmeasured model, human, runtime, domain, training, and novelty claims visibly withheld?
12. Would the paper still be informative if the strongest baseline matches the witness? It should.